

PICOPATT Internship

Schedule & Deliverables Update

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Simulated dataset (data/picoclimate_test): how it was generated

- Simulate **12 locations** across **Nantes** and **Montpellier** over **30 days**.
- Collect **4 time slots/day** (06/12/18/23 UTC) and **23 variables** per slot (weather + environment proxies).
- Add realistic missing data: some missing values + occasional full-sensor block outages (`missing_block_flag`).
- Produce two tables: `raw_measurements.csv` (slot-level) and `window_features.csv` (daily features for clustering).

Dataset samples (first 5 rows, selected columns)

raw_measurements.csv

	timestamp	location_id	time_slot	air_temp_c	rel_humidity_pct	solar_wm2	traffic_index	missing_block_flag	wind_speed_ms	sky_view_factor	co2_ppm	no2_ppb	o3_ppb	noise_db	traffic_index	pedestrian_index
0	2025-09-01 06:00:00+00:00	loc_000	morning	-15.0	67.2	0.0	0.857	0	NaN	0.516913	1152.0	174.466025	0.0	NaN	0.857	0.848
1	2025-09-01 12:00:00+00:00	loc_000	noon	-15.0	58.6	164.0	NaN	1	15.004	0.516913	NaN	NaN	NaN	NaN	NaN	NaN
2	2025-09-01 18:00:00+00:00	loc_000	evening	-15.0	85.7	183.0	0.791	0	25.000	NaN	408.0	NaN	0.0	110.0	0.791	1.000
3	2025-09-01 23:00:00+00:00	loc_000	night	-15.0	100.0	581.0	0.536	0	25.000	0.516913	637.0	NaN	0.0	110.0	0.536	1.000
4	2025-09-02 06:00:00+00:00	loc_000	morning	-15.0	87.4	0.0	0.831	0	21.630	0.516913	2033.0	175.178458	0.0	105.9	0.831	0.727

window_features.csv

	location_id	date	present_fraction	traffic_index_mean	water_proximity_mean	water_proximity_std	water_proximity_trend	heat_index_c_mean	heat_index_c_std	heat_index_c_trend	true_regime
0	loc_011	2025-09-26	0.923913	0.640750	0.052023	8.498375e-18	-3.301594e-34	-19.795000	0.410000	-0.082000	mixed_background
1	loc_011	2025-09-25	0.956522	0.611250	0.052023	0.000000e+00	0.000000e+00	-19.940000	0.120000	0.072000	mixed_background
2	loc_011	2025-09-24	0.913043	0.458667	0.052023	0.000000e+00	0.000000e+00	-19.860000	0.280000	-0.056000	mixed_background
3	loc_011	2025-09-23	0.891304	0.747667	0.052023	0.000000e+00	0.000000e+00	-19.642500	0.715000	-0.143000	mixed_background
4	loc_000	2025-09-08	0.923913	0.806250	0.000000	0.000000e+00	0.000000e+00	-19.373333	1.085419	-0.134286	mixed_background

Defense + submission (key dates)

Jun 26

Report target

Submit report

Jul 2

Upload slides

24 hours before the defense

**Jul 3
11:45**

Defense (in person)

15-min presentation

- June 8: first report draft (v0.1).
- June 26: evaluation form deadline (lab supervisors).
- June 27 (midnight AoE): university deadline.
- Slides: 15 minutes (aim 1 slide per minute).
- Report format: scientific paper in LaTeX, $\geq 12-14$ double-column pages.

Work plan (now → June)

- Start writing the final report now; lock structure early.
- Implement the changes we agreed last meeting while generating final figures.
- Maintain reproducible outputs to support report tables/plots.
- Build a 15-min defense deck in parallel once the report narrative stabilizes.

After the defense

- After July 3: continue internship normally at the lab.
- Paper/manuscript submission: discuss scope and timing after the defense.

Thanks — questions / suggestions welcome.